

Saruchana Yongchaiwathana

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SUMMARY

Data scientist with 4+ years of industry experience in pricing optimization, experimentation and forecasting, now completing an MSc in Informatics (AI) at the University of Zurich. I build models that reach production, from survival models and reinforcement-learning pricing controls to statistical tests for A/B testing platforms, and more recently LLM agent systems with evaluation and guardrails.

EDUCATION

Master of Science in Informatics **Sep 2025 - Present**
University of Zurich, Switzerland

- Cumulative GPA: 5.23/6.00 (excluding German language coursework)
- Major: Artificial Intelligence, Minor: Finance
- **Master's Project: Disentangling Decision-Making with disLRUs:** Modelled behavioural sequences from three decision-making algorithms (Sticky, Win-Stay-Lose-Shift, Q-learning) with disentangled Linear Recurrent Units (the lab's unpublished disLRU architecture) in JAX/Flax. Showed interpretable, algorithm-specific latent structure; detected switches between algorithms with multiple-stack model whose stacks specialize to distinct algorithms without supervision.

Bachelor of Engineering (Computer Engineering) - First Class Honours **Aug 2017 - May 2021**
Sirindhorn International Institute of Technology, Thammasat University, Thailand

- Cumulative GPA: 3.93/4.00
- **Young Scientist and Technologist Programme (YSTP) Scholarship** (NSTDA, Aug 2017 – May 2021): fully funded 4-year national scholarship (tuition + stipend).

WORK EXPERIENCE

Senior Data Scientist - Agoda Services Co., Ltd., Thailand **Apr 2022 - Aug 2025**
Website: https://www.agoda.com/about-agoda/

Pricing Optimization team:

- **Survival Analysis & Cost Forecasting:** Reframed cashback-claim prediction as a hazard-rate survival model (LightGBM, chronological train/test design, resampling with weight correction), after decomposing the claim rate to show that behaviour shifts made the old 180-day training lag a real accuracy problem. Cut the training lag from 180 to 60 days, enabled daily updates and removed a more-than-20% cost overestimate during the claim window.
- **Soft-Cap Discount Framework:** Replaced sequential hard caps on discount and cashback, whose insensitive regions hid the RL pricing model's exploration, with a simultaneous soft cap that rescales both levers within the shared margin. Tuned it by grid search, validated convergence by simulating the full RL loop and led alignment with engineering and business teams before launch; across ~6.6M production bookings, more than double the cashback treatment became readable as signal.

Experimentation team:

- **Interaction Between Experiments:** Derived a correct statistical test for interaction between concurrent A/B tests, accounting for the covariance from users in several cells, with a fast mode for routine screening and an exact per-user mode. Simulated both on real booking distributions and integrated them into the experimentation platform; the fast mode is reliable above roughly 1,000 users per group.
- **Metric Signal-to-Noise Study:** Measured which A/B test metrics are both sensitive (p-value distributions across historical experiments) and predictive of incremental bookings, decomposing composite metrics layer by layer. Showed that unique booker count often beats bookings per user as a primary metric, and that behavioral metrics are sensitive but weak predictors of business impact.

Associate Data Scientist & Data Scientist Intern - Sertis Co., Ltd., Thailand **Jan 2021 - Mar 2022**
Website: www.sertiscorp.com

- **Multi-Objective Optimization:** Extended and deployed on Google Cloud Platform a multi-objective steel-cutting model balancing yield loss, production cost and inventory cost; the optimized schedule achieved lower total cost than manually engineered plans.
- **Demand Forecasting:** Generalized the promotion-modelling component of an FMCG demand-forecasting pipeline to promotions of varying duration, improving predictive accuracy across enterprise client deployments.

Data Science Internship Bootcamp - KBTG (Kasikorn Business-Technology Group), Thailand **Jun 2020 - Jul 2020**

Website: kbtg.tech

- **Credit Modelling & Segmentation:** Built a credit-card propensity-to-revolve classification model and a spending-behaviour clustering model for campaign targeting; received the Best Business Analytics Award.

PROJECTS

job-radar: Agentic Job-Search Assistant *(GitHub, personal, 2026, ongoing)*

- **Screening & Evaluation:** 5-node LangGraph pipeline that turns raw postings into a ranked, explainable shortlist, scored by an evaluation harness against hand-labelled jobs (80% match-level agreement within one grade).
- **Structuring the Job Market:** LLM role cards and local BGE embeddings; job families learned from bootstrapped UMAP + HDBSCAN runs reconciled with an LLM-built taxonomy, with new jobs assigned at thresholds calibrated for 95% precision.
- **Grounded Application Material:** CVs and cover letters through a draft/critique loop grounded in a verified project database, with deterministic checks the LLM cannot override; FastAPI + HTMX dashboard used daily on 100+ real postings.

PearlSpinningTop: Conversational Movie Assistant *(GitHub, course project, UZH, HS2025, 2-person team)*

- **LLM Agent Harness & RAG:** Built the factual question-answering pipeline: FAISS retrieval of predicates and SPARQL examples as few-shot context for a local LLM (Qwen3-4B via Ollama), SPARQL execution against an RDF knowledge graph, and repair-and-retry on failure.
- **Routing & Recommendation:** Built the query router that dispatches each message to the right module, and a hybrid recommender combining item-item collaborative, knowledge-graph genre and Sentence-Transformer metadata similarity; evaluated in 3 live course evaluation sessions.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Scala, Java, TypeScript
- **Data Engineering & Visualization:** pandas, NumPy, Spark, Hadoop, Tableau, dashboarding
- **Machine Learning & AI:** scikit-learn, LightGBM, CatBoost, TensorFlow, JAX, PyTorch, SHAP
- **LLM & NLP:** LangGraph, LLM API, Ollama, RAG, LLM agent harnesses, Pydantic, FAISS, knowledge graphs, prompt engineering
- **Statistics & Experimentation:** A/B testing, experiment design, hypothesis testing, simulation
- **MLOps & Engineering:** MLflow, Kubeflow Pipelines, CI/CD, automated testing, model monitoring & alerting, Google Cloud Platform

LANGUAGES

- **Thai:** Native
- **English:** Professional Proficiency
- **German:** A2 (taking language course at UZH Sprachenzentrum)

VOLUNTEERING & INTERESTS

- **Volunteering:** Secretary, Association of Thai Students in Switzerland (ATSS), 2026: event organisation, accounting, meeting notes, administration.
- **Specialty coffee:** creator of the Wann_4 YouTube channel; barista and latte art competitor.
- **Sports:** skiing, badminton, dance, running.